

COMMON COURSE ASSIGNMENT FORMAT

Deep Learning-Based Deepfake Detection for Social Media Content

Assignment Information

Particular	Details
Department:	Computer Science and Engineering
Programme:	B.Tech (Computer Science and Engineering)
Course Code & Course Name	CS8XX – Deep Learning
Academic Year / Batch	2026–27 / VII Semester
Faculty Name	Dr.k.Krishnakumar
Student Name	A.Tharun Kumar Reddy
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Assignment Title	Deep Learning-Based Deepfake Detection for Social Media Content
Date of Issue	[03/09/2026]
Date of Submission	[03/09/2026]
Maximum Marks	100
Course Outcome(s) – CO	CO4: Design and evaluate deep learning models for real-world classification problems
Bloom's Taxonomy Level	L4 – Analyze, L5 – Evaluate, L6 – Create
SDG Mapping (SDG 1-17)	SDG 9 (Industry, Innovation & Infrastructure), SDG 16 (Peace, Justice & Strong Institutions)

Particular	Details
Industry / Societal Relevance	Combating misinformation and identity fraud on social media platforms through AI-based content authenticity verification

Assignment Problem / Challenge

This assignment presents a realistic, industry-relevant computing problem that requires students to apply deep learning concepts to detect AI-generated (deepfake) images and videos circulating on social media platforms, rather than reproduce textbook theory.

The assignment requires students to:

- Apply course concepts (CNNs, transfer learning, model evaluation) rather than reproduce theory.
- Identify and analyze the deepfake detection problem in the context of social media content moderation.
- Consider requirements and constraints such as accuracy, inference speed, dataset bias and adversarial robustness.
- Develop or compare alternative deep learning architectures (e.g., CNN-based vs. transformer-based detectors).
- Use relevant modern computing tools such as Python, TensorFlow/PyTorch and cloud/GPU platforms.
- Interpret results using appropriate performance metrics.
- Make and justify engineering decisions regarding model choice and deployment trade-offs.
- Consider technical, ethical, societal and safety factors relevant to misinformation and privacy.

Problem Statement

Problem / Case / Design Challenge:

A social media platform has observed a sharp rise in manipulated videos and images ("deepfakes") being uploaded and shared by users, several of which have been used to spread misinformation and impersonate public figures. The platform's trust-and-safety team has asked your team to design and prototype a deep learning-based system that can automatically flag likely deepfake images/videos uploaded to the platform, before they are surfaced widely to other users.

You are provided with a labelled dataset of real and fake face images/video frames (e.g., a subset of a public deepfake-detection dataset such as FaceForensics++ or a similar Kaggle deepfake dataset). Using this data, you are required to design, implement, evaluate and critically analyze a deep learning pipeline that classifies content as "authentic" or "manipulated", and to discuss how such a system could be deployed responsibly at social-media scale.

Requirements and Constraints

Students shall identify or be provided with the relevant requirements and constraints. For this assignment, the following are particularly relevant:

- Functional requirement: classify an uploaded image/video frame as "real" or "fake" with a confidence score.
- Performance requirement: target classification accuracy / F1-score and acceptable false-positive rate for flagging genuine content.
- Technical constraints: limited labelled training data; variation in lighting, resolution, compression and face angle across social media uploads.
- Cost, Time: inference must run within near-real-time limits during content upload/moderation review.
- Resources: availability of GPU/cloud compute for training and limited compute at inference (edge/server-side moderation).
- Reliability: robustness to unseen deepfake generation methods not present in the training data (generalization).
- Sustainability: computational/energy cost of training and repeatedly running large models at platform scale.
- Environmental and Ethical considerations: risk of false accusations against genuine users; bias across skin tone, gender and ethnicity in detection accuracy.
- Security/privacy: handling of user-uploaded facial data in compliance with data protection norms.

Only constraints relevant to this course/problem need to be considered.

Student Work

Problem Understanding and Formulation

Explain:

- What is the problem? (Automated detection of manipulated/synthetic facial content in social media uploads.)
- What are the expected outcomes? (A trained classifier with acceptable accuracy and a discussion of its deployability.)
- What information/data is available? (Labelled real/fake image or video-frame dataset, class distribution, sample counts.)
- What assumptions are made? (e.g., faces are detectable/cropped, dataset is representative, labels are correct.)
- What constraints must be satisfied? (Accuracy/F1 target, inference-time limits, fairness across demographic groups.)

Application of Course Knowledge

Identify and apply the relevant:

- Principles of convolutional neural networks (CNNs) and/or vision transformers for image classification.
- Transfer learning using pretrained backbones (e.g., XceptionNet, EfficientNet, ResNet).
- Loss functions (binary cross-entropy), regularization (dropout, data augmentation) and optimization algorithms.
- Evaluation methodology: train/validation/test split, k-fold cross-validation where appropriate.

Show the necessary calculations, derivations or reasoning — e.g., parameter counts, receptive field, or expected training time — to justify architectural choices.

Solution / Design / Methodology

Develop the proposed solution. This should include:

- Data pipeline: face extraction/cropping, resizing, normalization and augmentation.
- Model architecture: a CNN-based deepfake classifier (baseline) compared against transfer-learning models (XceptionNet, EfficientNet-B0) and a Vision Transformer (ViT) that treats face image patches as a sequence and uses self-attention to capture manipulation artifacts.
- Training methodology: hyperparameters, optimizer, learning-rate schedule, epochs, early stopping.

- Where appropriate, consider more than one possible solution — for example, a lightweight CNN suited to fast moderation versus a deeper transfer-learning model or a Vision Transformer suited to higher accuracy but slower inference — and compare them.

Proposed Deepfake Detection Pipeline

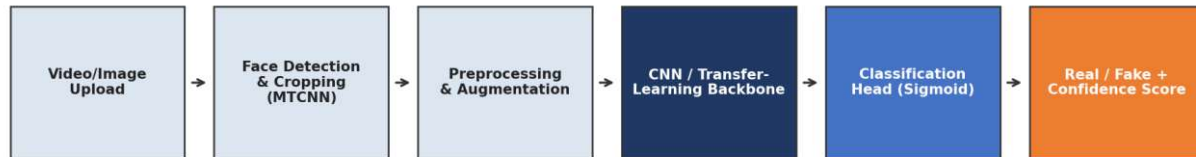


Figure 1: Proposed end-to-end deepfake detection pipeline, from upload to classification.

Use of Modern Tools

Use appropriate tools wherever relevant, for example:

CSE/IT: Python, TensorFlow/Keras or PyTorch, OpenCV/MTCNN for face detection, Google Colab/cloud GPU platforms, Git for version control, matplotlib/seaborn for result visualization.

Students shall provide appropriate evidence of tool usage and outputs (e.g., training logs, screenshots of confusion matrices, code repository link).

Results and Validation

Note: The figures below are illustrative sample outputs showing the type of evidence expected (students must replace these with their own generated results from their actual model runs).

Present the results using appropriate:

- Tables of accuracy, precision, recall and F1-score for each model tried.
- Graphs of training/validation loss and accuracy curves.
- Confusion matrix and ROC/AUC curve as simulation/experimental outputs.
- Screenshots of sample correctly and incorrectly classified images.
- Statistical comparison of the baseline CNN vs. the transfer-learning model.

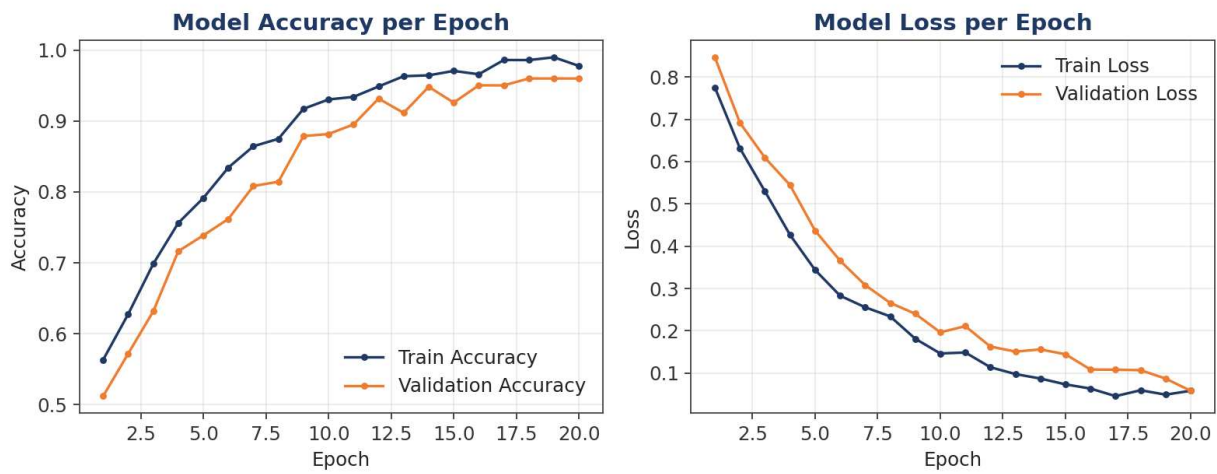


Figure 2: Training vs. validation accuracy and loss over 20 epochs (sample run).

**Confusion Matrix — Transfer-Learning Model
(Test Set, n = 1000)**

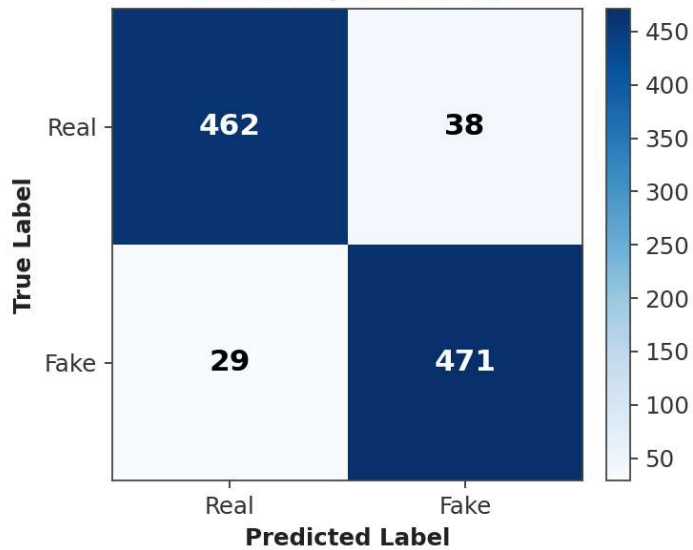


Figure 3: Confusion matrix of the transfer-learning model on the held-out test set.

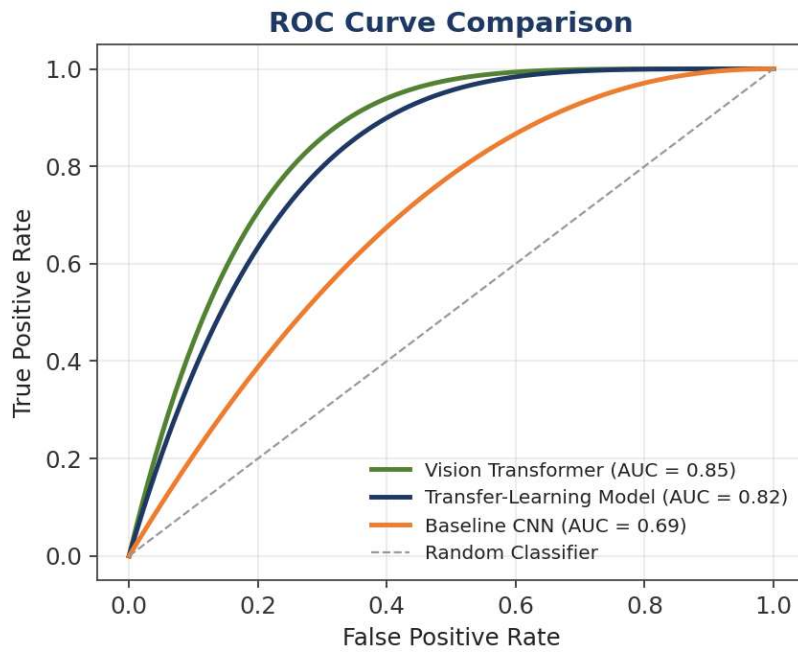


Figure 4: ROC curves comparing the baseline CNN, transfer-learning model and Vision Transformer.

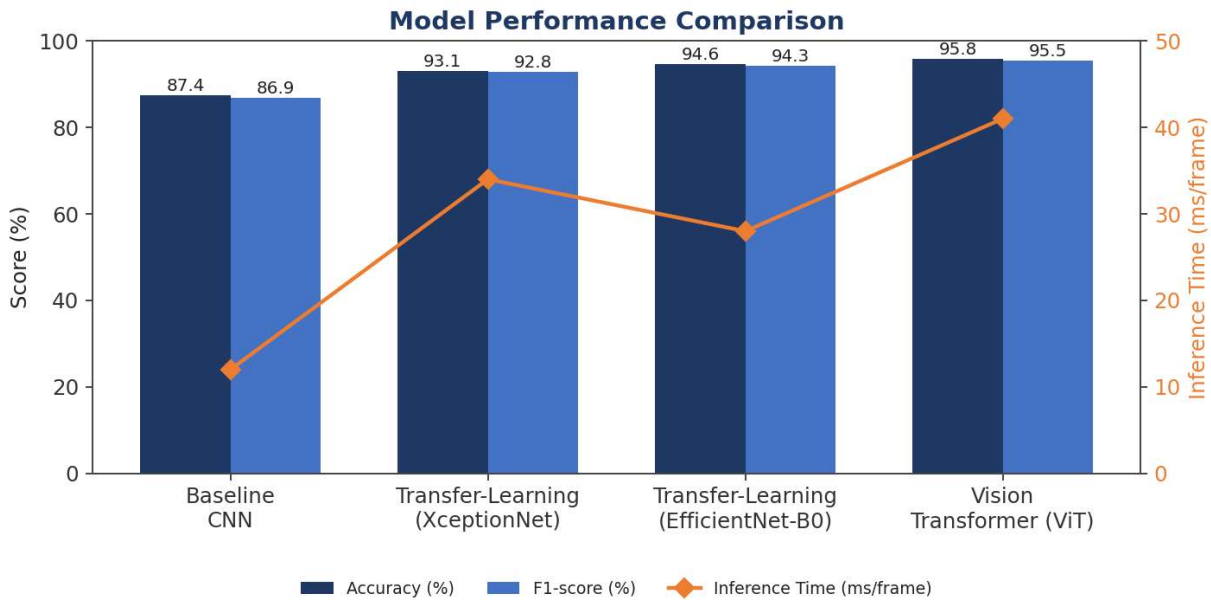


Figure 5: Accuracy, F1-score and inference time across four candidate algorithms.

Validate whether the proposed solution satisfies the stated accuracy, false-positive rate and inference-time requirements.

Analysis and Engineering Decision

Interpret the results. Where applicable:

- Compare the baseline CNN, transfer-learning models and the Vision Transformer on accuracy vs. inference speed.
- Identify advantages and limitations of each approach (e.g., generalization to unseen manipulation techniques, ViT's higher compute cost vs. its stronger attention-based artifact detection).
- Discuss trade-offs between model complexity, accuracy and deployability at social-media scale.
- Explain why the final model was selected for the proposed deployment scenario.
- Justify the decision using the quantitative evaluation metrics obtained.

Broader Considerations

Where relevant to the problem, discuss the impact of the proposed solution with respect to:

- Society: mitigating the spread of misinformation and protecting public discourse.
- Ethics: risk of wrongly flagging genuine content and consequent harm to users.
- Safety: preventing impersonation-based fraud and harassment.
- Accessibility and fairness: ensuring comparable detection accuracy across demographic groups.
- Professional responsibility: transparency in how flagged content is reviewed and appealed.

Conclusion

Summarize:

- The proposed deepfake detection solution and its design rationale.
- Major findings from model training and evaluation.
- The extent to which the stated requirements were achieved.
- Limitations of the current solution (e.g., dataset size, generalization gaps).
- Possible improvements (e.g., ensemble methods, temporal/video-level analysis, larger and more diverse datasets).

Student Reflection

- What did you learn from this assignment beyond what was directly taught in the classroom?
- If you were given additional time/resources, what would you improve and why?

References

Use an appropriate citation format (e.g., IEEE) and acknowledge all external sources, datasets (e.g., FaceForensics++, DFDC), standards, software libraries and any AI-assisted tools used in completing this assignment.

Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge (deep learning principles)	20
Solution methodology / model design / implementation	20
Use of appropriate modern tools / techniques (Python, TensorFlow/PyTorch, etc.)	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility	5
Technical documentation & reflection	5
TOTAL	100

CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO4	PO1, PO2	L3/L4	10

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Application of knowledge	CO4	PO1, PO2	L3/L4	20
Solution / Design / Implementation	CO4	PO3, PO5	L4/L5/L6	20
Modern tool usage	CO4	PO5	L3/L4	10
Validation	CO4	PO4	L4/L5	15
Analysis & justification	CO4	PO2, PO4	L4/L5	15
Broader considerations	CO4	PO8, PO12	L4/L5	5
Documentation & reflection	CO4	PO10	L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by this assignment. Every assignment is not required to map to every PO.

Faculty Evaluation & Continuous Improvement — CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$